

AI/ML Engineering Question Bank

100 scenario-based interview questions with answer frames

Prepared for Pune / Remote India AI-ML Engineering interviews

Basic + practical + production-grade knowledge in one structured set

How to use this PDF

Read for understanding first, then revise using the checklists and interview answer frames.

For each concept, practice saying: problem, data, model, evaluation, deployment, monitoring, failure handling.

This is written to reduce bookish answers and build production language for interviews.

Question Bank Structure

This bank contains 100 scenario-based and conceptual questions. Practice answering each in two minutes using: definition, production risk, metric, monitoring and trade-off.

Answer component	What to say
Definition	Explain the concept in one simple sentence.
Practical use	Mention the real use case or workflow.
Metric	Name how you would measure success.
Production risk	State one realistic failure mode.
Control	Give monitoring, fallback or rollback.

Python and Data

Q1. How do you structure a production ML Python project?

Strong answer structure

- Direct production answer: Separate src, configs, tests, notebooks, data contracts and deployment files.
- Use requirements/poetry, environment variables and reproducible seeds.
- Add logging, unit tests and README architecture.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q2. How do you handle missing values in a dataset?

Strong answer structure

- Direct production answer: Check whether missingness is random or meaningful.
- Use imputation only after preserving missing flags where useful.
- Fit imputers on training data only.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q3. What is schema drift and how do you detect it?

Strong answer structure

- Direct production answer: Compare incoming columns, types and allowed values against a stored contract.
- Fail fast or route to quarantine if required fields change.
- Alert data owner before model quality drops.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q4. How do you validate data before model inference?

Strong answer structure

- Direct production answer: Validate schema, nulls, ranges, duplicates, freshness and business rules.
- Log rejected records and sample failures.
- Do not run inference on untrusted data.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q5. How do you design logging for an ML API?

Strong answer structure

- Direct production answer: Log request id, model version, feature version, latency, prediction and errors.
- Avoid logging raw PII unless policy allows.
- Use logs for debugging and monitoring.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q6. How do you avoid hardcoded configuration?

Strong answer structure

- Direct production answer: Use YAML/env vars/secret manager for paths, thresholds and model names.
- Version configs with deployments.
- Separate dev/test/prod config.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q7. How do you test feature engineering code?

Strong answer structure

- Direct production answer: Unit test each transformation.
- Use sample rows with expected outputs.
- Test leakage boundaries and missing data.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Q8. How do you read large CSV/Excel files efficiently?

Strong answer structure

- Direct production answer: Use chunking, Parquet, DuckDB/Polars or database loading.
- Avoid loading huge files fully into memory.
- Validate schema on samples and full aggregates.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Classic ML

Q9. Explain bias-variance trade-off with an example.

Strong answer structure

- Direct production answer: High bias underfits; high variance overfits.
- Use learning curves and validation gap.
- Fix with more features/model complexity or regularization/data.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q10. When is accuracy misleading?

Strong answer structure

- Direct production answer: Accuracy fails on imbalanced datasets.
- Use precision, recall, F1, PR-AUC and cost-based thresholds.
- Always compare confusion matrix.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Q11. How do you handle class imbalance?

Strong answer structure

- Direct production answer: Use stratified split, class weights, resampling and threshold tuning.
- Evaluate PR-AUC/recall/precision, not only accuracy.
- Check business capacity for false positives.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q12. How do you choose between logistic regression and XGBoost?

Strong answer structure

- Direct production answer: Logistic regression is simple, interpretable and strong baseline.
- XGBoost handles nonlinear tabular patterns well.
- Choose based on metric, latency, explainability and maintenance.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q13. How do you prevent data leakage?

Strong answer structure

- Direct production answer: Ask whether the feature exists at prediction time.
- Use time/group-aware split and pipeline transforms.
- Remove target-derived columns.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q14. What is cross-validation and when can it be wrong?

Strong answer structure

- Direct production answer: CV estimates generalization, but random CV fails for time-series/group leakage.
- Use time-series or group CV when needed.
- Keep final test untouched.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q15. How do you tune model thresholds?

Strong answer structure

- Direct production answer: Choose threshold based on business cost, not default 0.5.
- Plot precision-recall and confusion matrix by threshold.
- Revisit after production feedback.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q16. What do you monitor after deploying a classification model?

Strong answer structure

- Direct production answer: Track input drift, prediction distribution, latency, errors and delayed labels.
- Monitor by segment, not only global metric.
- Alert on quality and data anomalies.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q17. How do you explain a model to business users?

Strong answer structure

- Direct production answer: Use business language, top drivers and examples.
- Use SHAP/permutation importance carefully.
- Explain limitations and decision boundaries.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q18. What is model calibration and why does it matter?

Strong answer structure

- Direct production answer: Calibration means predicted probabilities match real frequencies.
- Useful when probability drives decisions.
- Check calibration plots and Brier/log loss.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Deep Learning and NLP

Q19. Explain backpropagation simply.

Strong answer structure

- Direct production answer: Forward pass predicts, loss measures error, backward pass computes gradients.
- Optimizer updates weights to reduce loss.
- Learning rate controls stability.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q20. Why do transformers work well for language?

Strong answer structure

- Direct production answer: Transformers use attention to connect tokens directly.
- They train in parallel and capture long-range context better than RNNs.
- They power modern LLMs and embedding models.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q21. What is attention?

Strong answer structure

- Direct production answer: Attention scores how much each token should focus on other tokens.
- Self-attention builds context-aware token representations.
- Multi-head attention learns different relationships.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q22. What are embeddings?

Strong answer structure

- Direct production answer: Embeddings are numeric vectors representing meaning.
- Similar items should be close in vector space.
- Used in semantic search, clustering and recommendations.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q23. How does cosine similarity work in semantic search?

Strong answer structure

- Direct production answer: Cosine similarity measures angle between vectors.
- Useful when direction matters more than magnitude.
- Common in embedding retrieval.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q24. What is the difference between BERT and GPT-style models?

Strong answer structure

- Direct production answer: BERT is encoder-style and strong for understanding tasks.
- GPT-style models are decoder/generative and strong for text generation.
- Choose based on task.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q25. How do you choose an embedding model?

Strong answer structure

- Direct production answer: Evaluate on domain queries and documents.
- Check language support, dimensions, speed and cost.
- Version embeddings because changing model requires reindexing.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q26. What is tokenization and why does it matter?

Strong answer structure

- Direct production answer: Tokenization converts text to model units.
- Affects context length, cost and multilingual behavior.
- Long inputs must be chunked or summarized.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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LLMs and Prompting

Q27. Why do LLMs hallucinate?

Strong answer structure

- Direct production answer: LLMs generate likely text, not guaranteed truth.
- They may lack data, retrieve wrong context or overgeneralize.
- Use grounding, citations and abstention.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Q28. How do you reduce hallucination in production?

Strong answer structure

- Direct production answer: Use RAG, structured prompts, citations, validation and low temperature.
- Ask the model to abstain when evidence is missing.
- Evaluate faithfulness on golden examples.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Q29. What is the difference between temperature and top-p?

Strong answer structure

- Direct production answer: Temperature controls randomness distribution sharpness.
- Top-p samples from cumulative probability mass.
- Lower values improve consistency for production.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q30. What is structured output and why is it useful?

Strong answer structure

- Direct production answer: Force JSON/schema to make output machine-readable.
- Validate with Pydantic or JSON schema.
- Retry or repair invalid outputs.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q31. How do you handle prompt injection?

Strong answer structure

- Direct production answer: Treat user and retrieved text as untrusted.
- Use least-privilege tools and output validation.
- Separate instructions from data and add human approval for actions.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q32. How do you version prompts?

Strong answer structure

- Direct production answer: Store prompt templates with version, owner, tests and deployment history.
- Run regression tests before changing prompts.
- Log prompt version in traces.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q33. When would you use a local LLM?

Strong answer structure

- Direct production answer: Use local LLM for privacy, cost control or offline demos.
- Benchmark quality and latency on your hardware.
- Do not expose local inference endpoint publicly.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q34. How do you control LLM cost?

Strong answer structure

- Direct production answer: Cache repeated answers, route simple tasks to small models/rules, limit context and max tokens.
- Monitor cost per workflow.
- Use retrieval compression and batching where possible.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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RAG

Q35. Explain RAG end to end.

Strong answer structure

- Direct production answer: Ingest and parse documents, chunk with metadata, embed, index, retrieve, rerank, generate with citations and evaluate.
- Monitor retrieval and generation separately.
- Add abstention for weak evidence.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q36. How do you choose chunk size?

Strong answer structure

- Direct production answer: Choose based on document structure and evaluation, not guesswork.
- Too small loses context; too large adds noise.
- Track retrieval recall and answer faithfulness.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q37. What metadata do you store with chunks?

Strong answer structure

- Direct production answer: Store source, page, section, date, owner, permissions, document type and version.
- Metadata enables filtering, citations and access control.
- Missing metadata makes debugging hard.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q38. How do you evaluate a retriever?

Strong answer structure

- Direct production answer: Use golden queries with expected relevant chunks.
- Measure recall@k, MRR/nDCG and manual relevance.
- Analyze misses by document type and query style.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q39. What is hybrid search?

Strong answer structure

- Direct production answer: Combine keyword/BM25 with vector search.
- Useful for exact names, tickers, IDs and semantic intent.
- Tune merge/rerank strategy.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q40. What is reranking?

Strong answer structure

- Direct production answer: Reranker rescores candidate chunks for relevance.
- Improves precision at the cost of latency.
- Use after broad retrieval.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q41. What is Corrective RAG?

Strong answer structure

- Direct production answer: Grade retrieval quality before answering.
- Rewrite query or retrieve again if context is insufficient.
- Abstain or ask clarification if still weak.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Q42. How do you prevent unsupported citations?

Strong answer structure

- Direct production answer: Generate answer only from retrieved chunks and cite exact sources.
- Validate that each claim maps to evidence.
- Use post-generation citation checker.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q43. Why might RAG fail even when the document exists?

Strong answer structure

- Direct production answer: Parsing may fail, chunking may split context, metadata filter may exclude it, embedding may miss it or reranker may demote it.
- Debug trace from query to retrieved chunks.
- Add golden test for that case.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q44. How do you update a vector index when documents change?

Strong answer structure

- Direct production answer: Use document ids and embedding version.
- Upsert changed chunks and delete removed chunks.
- Rebuild index when embedding model or chunking changes.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Agentic AI and LangGraph

Q45. What is an AI agent?

Strong answer structure

- Direct production answer: Agent uses state, tools and decision logic to complete tasks.
- Production agents need bounded steps, permissions and traces.
- Use deterministic workflow where possible.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q46. When should you not use an agent?

Strong answer structure

- Direct production answer: Avoid agents for simple deterministic tasks.
- Avoid when risk is high and actions are irreversible without approval.
- Use workflows/rules for predictable paths.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Q47. Why use LangGraph instead of a simple chain?

Strong answer structure

- Direct production answer: LangGraph is useful for stateful, long-running, multi-step workflows.
- Nodes, edges and state make behavior debuggable.
- It supports persistence, interrupts and routing.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q48. What is state in LangGraph?

Strong answer structure

- Direct production answer: LangGraph is useful for stateful, long-running, multi-step workflows.
- Nodes, edges and state make behavior debuggable.
- It supports persistence, interrupts and routing.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q49. How do you avoid infinite loops in agents?

Strong answer structure

- Direct production answer: Set max steps, loop counters and stop conditions.
- Use confidence thresholds and human escalation.
- Trace loops and failed tools.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q50. How do you handle tool failure?

Strong answer structure

- Direct production answer: Validate inputs, catch recoverable errors and retry transient failures.
- Return tool error to state for recovery.
- Escalate after max attempts.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q51. How do you secure agent tool access?

Strong answer structure

- Direct production answer: Use least privilege, scoped credentials, allowlisted tools and audit logs.
- Validate tool arguments.
- Require approval for sensitive actions.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q52. What is human-in-the-loop and when is it required?

Strong answer structure

- Direct production answer: Use for low confidence, high financial/legal risk or irreversible action.
- Store human decision as feedback.
- Design review queues and SLAs.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q53. How do you evaluate an agent?

Strong answer structure

- Direct production answer: Measure task success, tool-call accuracy, step count, cost, latency and human override rate.
- Use scenario tests.
- Inspect traces for failure modes.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q54. How do you design memory for an agent?

Strong answer structure

- Direct production answer: Separate session state from long-term memory.
- Store only useful, consented facts.
- Use retrieval and deletion controls.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Fine-Tuning

Q55. RAG vs fine-tuning: how do you decide?

Strong answer structure

- Direct production answer: Use RAG for private or changing knowledge.
- Use fine-tuning for behavior, style or narrow task consistency.
- Compare both against baseline.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q56. What is LoRA?

Strong answer structure

- Direct production answer: LoRA trains low-rank adapter weights while base model stays mostly frozen.
- It reduces training cost and storage.
- Adapters are easier to version and swap.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q57. What is QLoRA?

Strong answer structure

- Direct production answer: LoRA trains low-rank adapter weights while base model stays mostly frozen.
- It reduces training cost and storage.
- Adapters are easier to version and swap.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q58. How do you prepare data for fine-tuning?

Strong answer structure

- Direct production answer: Clean instruction-output pairs, remove duplicates and bad labels.
- Include edge cases and refusal examples.
- Split train/validation/test and track data version.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q59. How do you know a fine-tuned model improved?

Strong answer structure

- Direct production answer: Use same golden set before/after tuning.
- Compare to base, prompt-only and RAG solution.
- Check regressions and safety.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q60. What is catastrophic forgetting?

Strong answer structure

- Direct production answer: Model loses previous capability after training on narrow data.
- Use diverse training data and evaluation across old/new tasks.
- Prefer adapters for safer specialization.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q61. How do you deploy a fine-tuned model?

Strong answer structure

- Direct production answer: Package adapter/base model, register version and serve via endpoint.
- Monitor quality, latency and cost.
- Keep previous model for rollback.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Evaluation and LLMOps

Q62. What is a golden dataset?

Strong answer structure

- Direct production answer: Curated representative examples with expected answers and scoring rules.
- Used for regression testing prompts/models/retrievers.
- Must include edge cases and failures.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q63. How do you evaluate RAG answer quality?

Strong answer structure

- Direct production answer: Evaluate retrieval relevance, answer faithfulness, answer relevance and citation accuracy.
- Use human review and LLM judge with rubrics.
- Track failure root causes.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q64. What is LLM-as-judge and what are its risks?

Strong answer structure

- Direct production answer: LLM judge scales evaluation but can be biased or inconsistent.
- Use clear rubric and calibration examples.
- Spot-check with humans.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q65. What should be included in an LLM trace?

Strong answer structure

- Direct production answer: Record prompt, model, parameters, retrieved context, tool calls, output, tokens, latency, cost and feedback.
- Mask sensitive data.
- Use trace for debugging.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q66. How do you track token cost?

Strong answer structure

- Direct production answer: Log input/output tokens per request, workflow and user.
- Alert on cost spikes.
- Optimize with routing, caching and context compression.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q67. How do you monitor latency?

Strong answer structure

- Direct production answer: Track p50/p95/p99 latency and time spent in retrieval, model and tools.
- Add timeouts and fallbacks.
- Optimize slow components.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q68. How do you create feedback loops?

Strong answer structure

- Direct production answer: Collect thumbs, corrections, root-cause labels and user outcome.
- Feed into evaluation dataset.
- Prioritize recurring failures.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q69. How do you do regression testing for prompts?

Strong answer structure

- Direct production answer: Run golden set before prompt release.
- Compare quality, safety, JSON validity and cost.
- Roll out with versioning.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q70. What alerts would you configure for a GenAI app?

Strong answer structure

- Direct production answer: Alert on error rate, timeout rate, cost spike, low judge score, retrieval failures, safety blocks and feedback drops.
- Separate noisy warnings from action alerts.
- Define owner.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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MLOps and Deployment

Q71. How do you deploy an ML model with FastAPI?

Strong answer structure

- Direct production answer: Load model at startup, validate request schema and return typed response.
- Add health endpoint and logging.
- Containerize and monitor.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q72. Why use Docker?

Strong answer structure

- Direct production answer: Docker makes dependencies reproducible.
- Build once and deploy consistently.
- Use slim images and pinned versions.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q73. What is a model registry?

Strong answer structure

- Direct production answer: Stores model versions, stages, metadata, metrics and approval status.
- Controls promotion and rollback.
- Improves governance.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q74. What is experiment tracking?

Strong answer structure

- Direct production answer: Tracks params, metrics, artifacts and code/data version.
- Enables reproducibility.
- Useful for comparing runs.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q75. How do you detect data drift?

Strong answer structure

- Direct production answer: Compare production feature distributions to training baseline.
- Use statistical tests and segment analysis.
- Confirm business cause before retraining.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q76. How do you rollback a bad model?

Strong answer structure

- Direct production answer: Keep previous model/prompt/index available.
- Use versioned deployments and traffic routing.
- Trigger rollback on quality, latency or error thresholds.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q77. Batch vs realtime inference: how do you choose?

Strong answer structure

- Direct production answer: Batch for scheduled high-volume scoring with relaxed latency.
- Realtime for user/API decisions with low latency.
- Sometimes combine both.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q78. What is canary deployment?

Strong answer structure

- Direct production answer: Send small traffic to new model first.
- Compare metrics against control.
- Rollback quickly if issues appear.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q79. How do you handle secrets in deployment?

Strong answer structure

- Direct production answer: Use secret manager or environment variables.
- Never commit keys.
- Rotate and scope permissions.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q80. How do you design CI/CD for ML?

Strong answer structure

- Direct production answer: Run code tests, data tests, model eval and packaging in pipeline.
- Promote only if metrics pass.
- Deploy with rollback.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

System Design and Scenarios

Q81. Design a RAG chatbot for company policy documents.

Strong answer structure

- Direct production answer: Use document ACLs, chunk metadata and cited answers.
- Add HR/legal review for sensitive policies.
- Monitor unanswered and wrong-answer feedback.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q82. Design an agentic RAG system for financial document analysis.

Strong answer structure

- Direct production answer: Combine PDF parsing, table extraction, RAG, SQL and validation.
- Cite page/table evidence.
- Use human review for investment-sensitive outputs.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q83. Design a support desk automation agent.

Strong answer structure

- Direct production answer: Classify ticket, retrieve KB, draft answer, create/update ticket and escalate low confidence.
- Track resolution rate and cost.
- Add human review for risky cases.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q84. Design a document extraction system for PDFs.

Strong answer structure

- Direct production answer: Parse layout, detect tables, extract fields to schema and validate against business rules.
- Return confidence and source page.
- Human review for low confidence.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q85. Design a churn prediction system.

Strong answer structure

- Direct production answer: Build features from customer behavior and history.
- Use time split and evaluate recall/precision.
- Connect predictions to retention workflow and outcome tracking.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q86. Your model accuracy is high offline but poor in production. What do you do?

Strong answer structure

- Direct production answer: Check train/serve skew, data drift, leakage, label delay and pipeline changes.
- Compare distributions and examples.
- Rollback if user impact is severe.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q87. Your RAG bot is slow. How do you reduce latency?

Strong answer structure

- Direct production answer: Measure retrieval, rerank, LLM and tool latency separately.
- Cache, reduce context, use smaller model or stream output.
- Optimize index and batch embeddings.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q88. Your LLM cost is increasing every month. What do you do?

Strong answer structure

- Direct production answer: Break down cost by user, workflow, model and token type.
- Add caching, routing, shorter context and budgets.
- Monitor cost per successful task.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q89. A user reports a wrong answer from your AI app. How do you debug?

Strong answer structure

- Direct production answer: Find trace, review query, retrieved chunks, prompt, model output and citations.
- Classify root cause.
- Add example to eval set.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q90. A document contains malicious instructions. How do you protect the system?

Strong answer structure

- Direct production answer: Treat document text as data, not instruction.
- Use prompt-injection filters, tool restrictions and output validation.
- Avoid tool execution from untrusted content.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

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Q91. The business asks for 99% accuracy. How do you respond?

Strong answer structure

- Direct production answer: Ask what error costs matter and whether classes are imbalanced.
- Translate to precision/recall/business metric.
- Set realistic baseline and evaluation plan.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q92. A new column appears in monthly data. How should the pipeline behave?

Strong answer structure

- Direct production answer: Pipeline should ignore unknown optional columns or quarantine if schema contract changes critically.
- Log and alert data owner.
- Update feature contract intentionally.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q93. Design a local LLM private assistant for internal documents.

Strong answer structure

- Direct production answer: Use local LLM for privacy, cost control or offline demos.
- Benchmark quality and latency on your hardware.
- Do not expose local inference endpoint publicly.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q94. Design an AI analytics dashboard that adapts to new monthly Excel data.

Strong answer structure

- Direct production answer: Build schema inference and validation, automatic feature engineering and dashboard refresh.
- Handle new rows and known columns dynamically.
- Quarantine broken rows and show data-quality report.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

Q95. Design a model monitoring dashboard for drift and quality.

Strong answer structure

- Direct production answer: Show drift, quality, latency, errors, prediction distribution and business KPIs.
- Allow segment drill-down.
- Trigger alerts and retraining review.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

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Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

LLMs and Prompting

Q96. How do you make LLM output deterministic enough for automation?

Strong answer structure

- Direct production answer: Use low temperature, structured output, schemas, validation and constrained choices.
- Use workflows/rules for critical decisions.
- Test outputs across runs.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.

RAG

Q97. What is the difference between context relevance and answer faithfulness?

Strong answer structure

- Direct production answer: Context relevance checks whether retrieved chunks are useful.
- Faithfulness checks whether answer is supported by chunks.
- Both are needed for RAG quality.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

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Agentic AI and LangGraph

Q98. How do you log every decision made by an agent?

Strong answer structure

- Direct production answer: Use traces with node name, state changes, tool inputs/outputs and routing decisions.
- Mask secrets and PII.
- Store enough for replay/debugging.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

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Security

Q99. What are the top LLM application security risks you consider?

Strong answer structure

- Direct production answer: Mention prompt injection, sensitive disclosure, supply chain, poisoning, improper output handling, excessive agency and unbounded cost.
- Use least privilege, validation and audit logs.
- Align with OWASP thinking.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

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Cloud

Q100. How would you deploy cheaply on AWS or GCP for an interview project?

Strong answer structure

- Direct production answer: Use small models, serverless/container free tiers, local embeddings and caching.
- Avoid GPUs unless needed.
- Track billing and auto-shutdown demo services.
- Also mention a baseline, an evaluation metric and a failure-handling plan.
- Close with business impact: time saved, risk reduced, cost controlled, quality improved or adoption increased.

Production keywords to include

baseline, data validation, evaluation metric, observability, latency/cost, fallback, rollback, security, business impact

Red flag to avoid

Do not answer only with a definition or only with library names. Interviewers want judgment, trade-offs and failure handling.